

Alexander Kamil Diab-Liu

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SUMMARY

Engineering student with hands-on experience in post-silicon validation, embedded systems, and test automation across high-performance GPUs and autonomous robotics. Passionate about scalable hardware-software integration and consumer-centric solutions driven by early exposure to marketing. Recognized for automation leadership at AMD and competitive hardware design at McMaster.

EDUCATION

McMaster University

April 2027

Bachelor of Electrical and Biomedical Engineering

Hamilton, ON

- **Relevant Courses:** Microprocessors, Logic Design, Electronic Devices & Circuits, Data Structures & Algorithms
- **Extracurriculars:** President (DISA McMaster), Executive (DeltaHacks, GDSC, MacSER, MacEngMusical)

PROFESSIONAL EXPERIENCE

Advanced Micro Devices (AMD)

May 2023 – August 2024

Software Engineering Intern (Data Center GPU Virtualization)

Markham, ON

- Received the AMD Intern Spotlight Award for designing a Python automation framework for post-silicon GPU virtualization testing, debug, and NPI, **reducing manual user intervention by 98%**
- Developed Bash, PowerShell, and Ansible scripts to automate Linux KVM and Microsoft Hyper-V VM deployment pipelines, improving test coverage and **reducing setup times by 80%**
- Conducted board-level debug and GPU failure analysis in post-silicon environments, **increasing test case coverage and hardware bring-up by 15%** through cross-functional debug sessions and root cause analysis
- Programmed 50+ automation workflows in Python and Bash to test **synthetic HPC and ML workloads** (Llama, GPT) across Linux and Windows, integrating with Dockerized validation infrastructure
- Led cross-team standardization of virtualization, RAS and Reset **test infrastructure development**, consolidating frameworks into a unified hardware-software validation suite adaptable across hypervisors and operating systems

PROJECTS

AI Smart Home Assistant | *Python, Raspberry Pi, AWS, Linux, SQLite3*

June 2025 - August 2025

- Built an AI-powered multimedia alarm system on a Raspberry Pi to reduce phone dependency and improve user well-being through voice control for alarms, music, and productivity functions
- Brought up a full voice assistant pipeline with I2C RTC hardware, Linux OS, and SQLite3, deploying NLP machine learning models for SVM intent classification and spaCy NER parameter extraction, achieving **96% accuracy**
- Modularized Python architecture and validated system behavior with Pytest unit testing, offloading Whisper transcription to **AWS EC2** for low-latency cloud inference and elasticity

3D Spatial Mapping System | *Embedded C, Python*

January 2025 - April 2025

- Engineered a compact, low-cost LiDAR alternative by integrating a VL53L1X Time-of-Flight sensor with an MSP432E401Y microcontroller (ARM Cortex-M4F), enabling 360° spatial scanning through GPIO motor control
- Implemented real-time I²C/UART communication and Open3D-based 3D visualization in Embedded C and Python, showcasing end-to-end embedded system integration, spatial data acquisition, and sensor fusion

Autonomous Car - McMaster Engineering Competition Finalist | *Arduino C++*

November 2024

- Designed and programmed an embedded autonomous vehicle using C++ and ultrasonic sensors, developing a modular obstacle detection and avoidance system processing multi-sensor input in real-time
- Engineered a motor control system with PWM-based speed tuning and directional switching, implementing real-world failure recovery and move-logging mechanisms to simulate field return diagnostics

TECHNICAL SKILLS

Programming & Scripting: Python, C/C++, Bash, Powershell, Ansible

Embedded Systems: Microcontrollers/Microprocessors, Oscilloscopes, Multimeters, Logic Analyzers, Raspberry Pi

Software/DevOps: Linux/Unix, Docker, ML/AI, Git, GitHub, GitLab, Cloud Computing (AWS), Version Control

Testing & Validation: Post-Silicon Verification, Automation, Test Plan Design/Development, Firmware Debugging, Hardware Bring-Up, Computer Architecture, Systems Design